FEMS MICRO MILAN 2025
Congress — Exhibition

Duke PRATT SCHOOL of ENGINEERING

# Optimizing Fungal Microencapsulation for Bioremediation of PAHs

**278.17P**

Erica Babusci and Claudia K Gunsch
*Department of Civil and Environmental Engineering, Duke University*
Erica.Babusci@duke.edu
(614)-600-6861

## Background

Polycyclic aromatic hydrocarbons (PAHs) are organic pollutants that are a significant public health concern due to their carcinogenic and mutagenic nature[1-3]. There are numerous strategies implemented to remediate PAH contamination, many being ecologically intrusive[4]. A sustainable and cost-effective alternative is **mycoremediation**. However, minimal success in PAH degradation has been seen due to planktonic microbes not properly establishing within the novel environment[5].

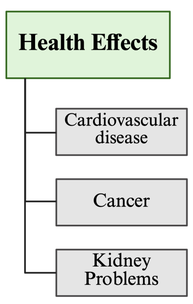



Figure 1: Health effects associated with PAH exposure

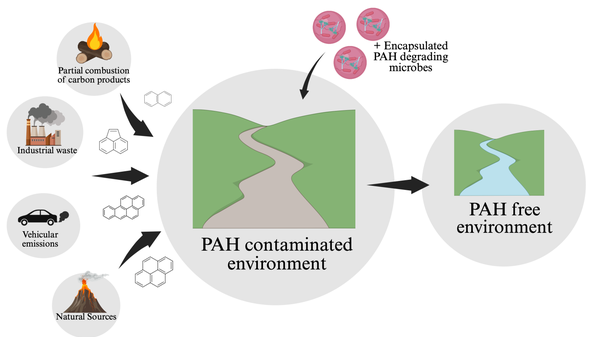



Figure 2: The general schematic of the application of capsules for in-situ bioremediation

To address this, we are **encapsulating PAH-degrading fungi in sodium alginate** to enhance their survival and degradation capacity in the environment. We are evaluating the difference between encapsulated and free-living cultures of environmental isolates of ***Trichoderma*** and ***Aureobasidium***[6] to measure growth, viability, and PAH degradation[7].

## Methods

### Capsule Formation

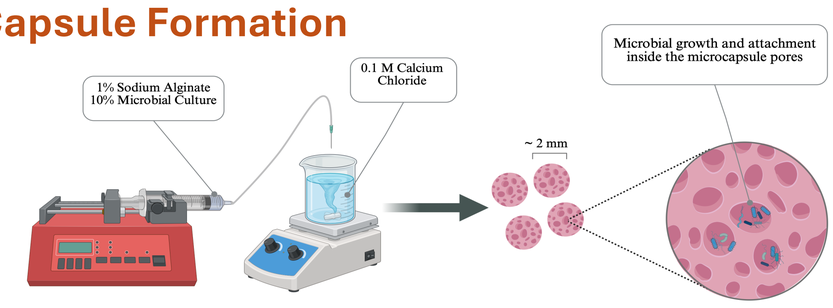



Figure 3: Syringe pump set up for forming capsules via extrusion[8]

### Different Capsules Being Tested

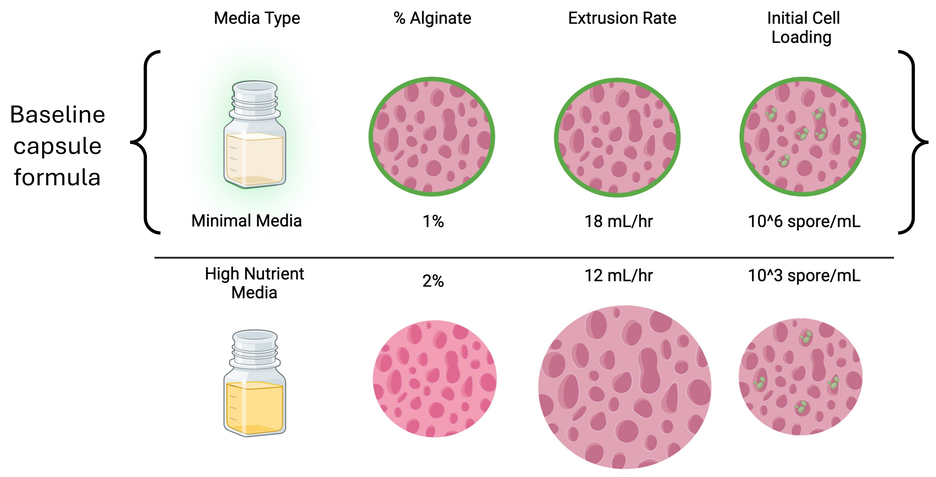



Figure 4: Different conditions in which capsules will be formed and compared to the baseline.

### Long Term Viability and PAH Degradation

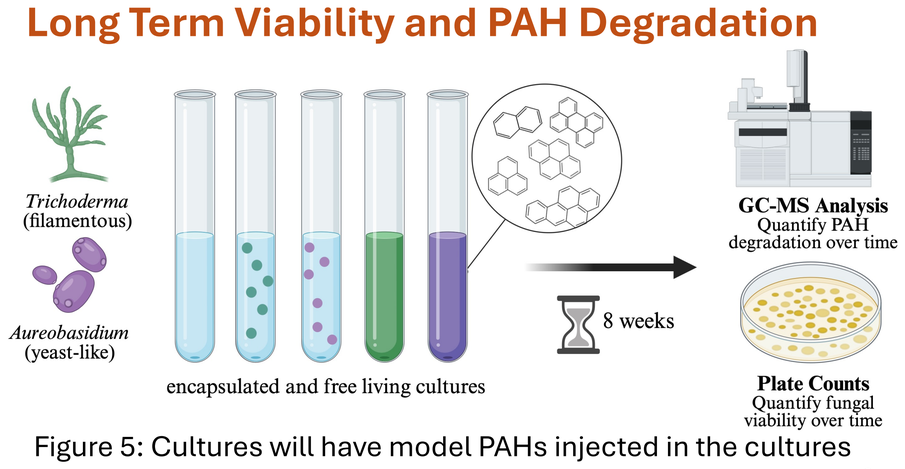



Figure 5: Cultures will have model PAHs injected in the cultures and let grow for 8 weeks, with data collected at 0,4, and 8 weeks[9].

FEMS MICRO Milan 2025: Congress & Exhibition

## Results

### Growth Curves Comparing Treatments

Trichoderma and Aureobasidium encapsulated cultures showed **shorter stationary phase growth** than free-living cultures for both extrusion rates but **Aureobasidium showed lower overall OD**

- 12 mL/hr promoted an earlier growth rate for both isolates

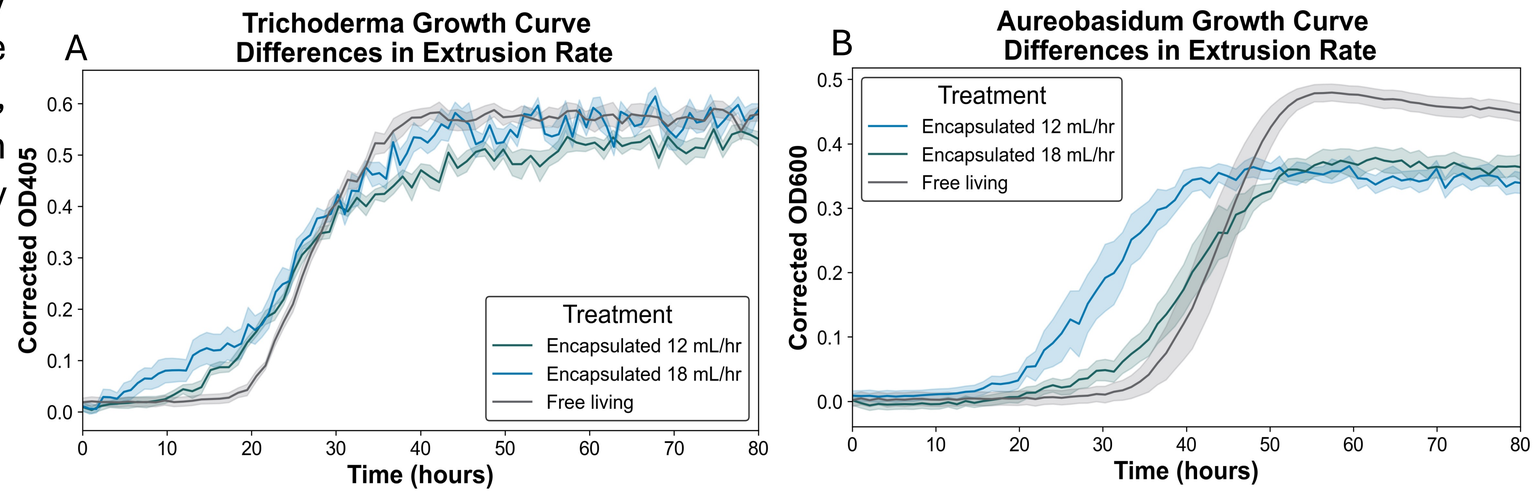



Figure 6 (A-B): Figures showing the corrected OD for (A) Trichoderma and (B) Aureobasidium over 80 hours, comparing capsules formed by extrusion rates of 12 mL/hr vs 18 mL/hr to free living cultures

Trichoderma and Aureobasidium encapsulated cultures showed **shorter stationary phase growth** than free-living cultures for both extrusion rates but **Aureobasidium showed lower overall OD**

- Alginate concentration does not affect encapsulated Trichoderma growth
- 2% alginate promoted earlier growth for Aureobasidium

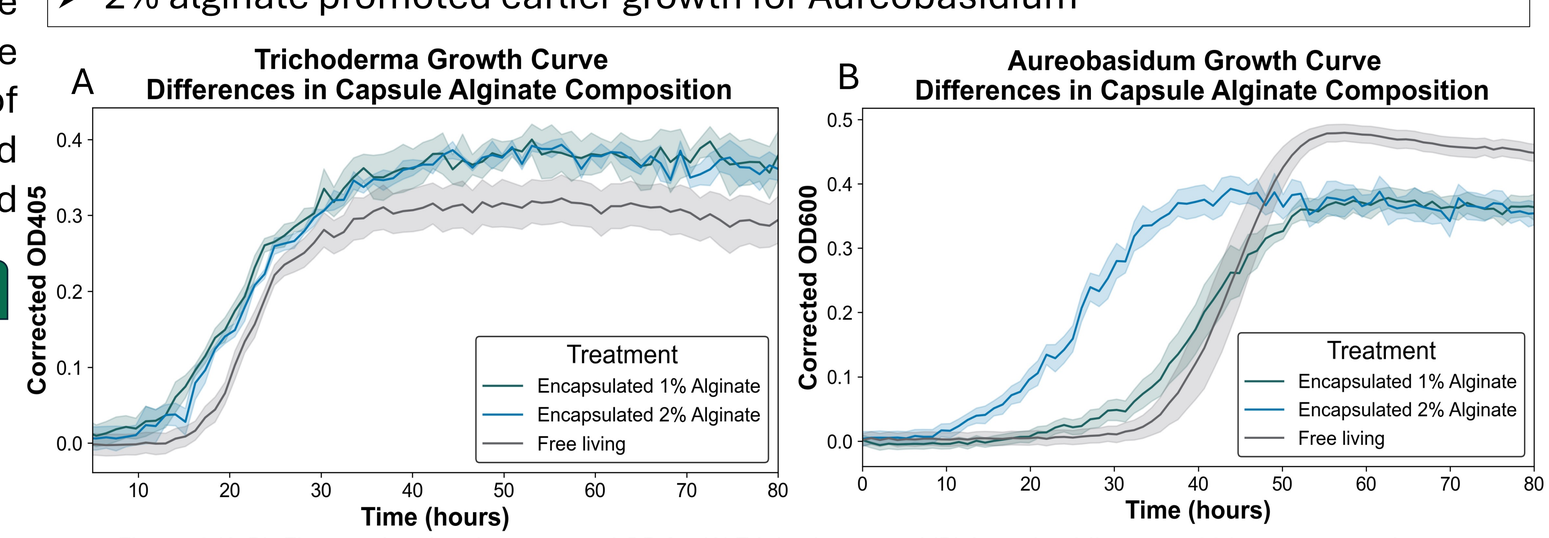



Figure 6 (A-B): Figures showing the corrected OD for (A) Trichoderma and (B) Aureobasidium over 80 hours, comparing capsules formed by 1% and 2% alginate to free living cultures

### Long Term Viability

Encapsulation **promoted viability of Trichoderma** cultures after 8 weeks but had **no effect on Aureobasidium**

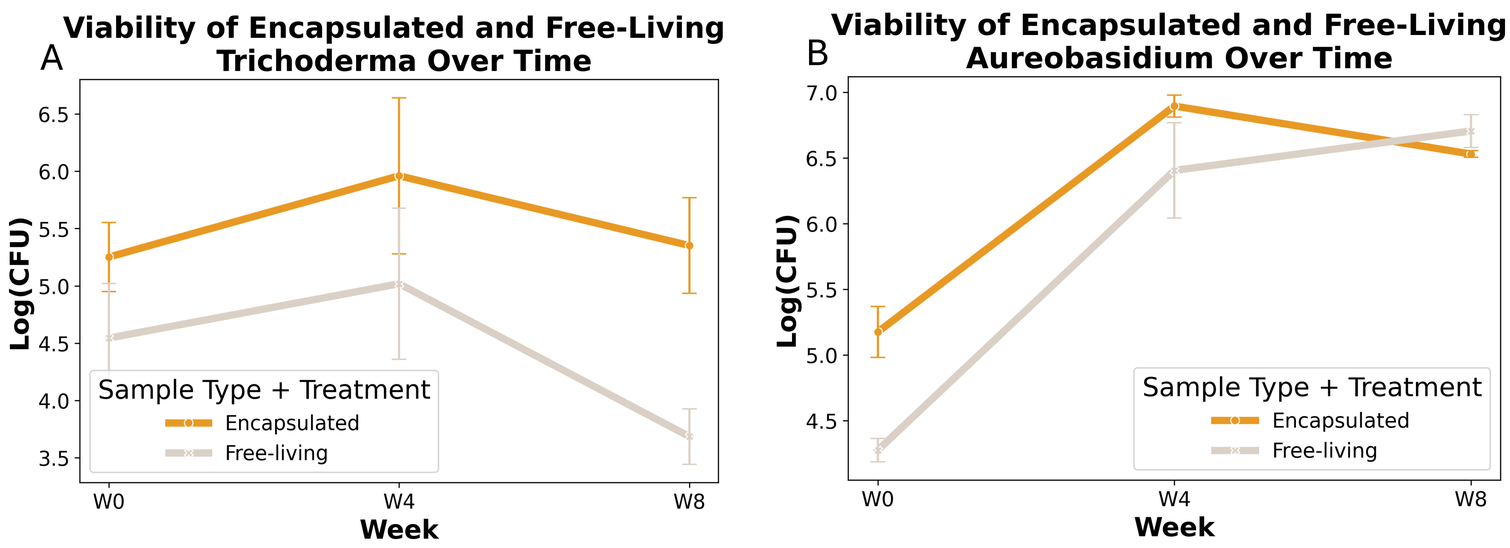



Figure 8 (A-B): Figures showing the total concentration of (A) Trichoderma and (B) Aureobasidium after 0,4, and 8 weeks. Capsuled formed with the baseline capsule formation

## References and Acknowledgments

[1] US EPA (2023), Superfund [2] Di Giulio, R. T., & Clark, B. W. (2015). *J Tox and Env Health* [3] Baek, et al. (1991) *Water Air Soil Pollut*. [4] Seo, J.-S., Keum, Y.-S., & Li, Q. X. (2009). Bacterial Degradation of Aromatic Compounds. [5] Omokhagbor Adams et al., (2020) *International Journal of Environmental Bioremediation & Biodegradation* [6] Valdivia-Rivera, et al. (2021). *Reviews in Environmental Science and Bio/Technology*. [7] Volkoff, et al. (2019) *Duke University*. [8] Yueng, et al. (2016) *Front. Microbial.* [9] Meletiadis, et al. (2001) *ASM Journals*

This material is based upon work supported by the National Institute of Environmental Health Sciences Under Grant Number P42ES010356

bio RENDER · Duke SUPERFUND Research Center · NSF · NIH NIEHSSRP